SYNTHETIC AUTOSAR HIGH-LEVEL DESIGN: ADAS CAMERA & RADAR FUSION

DOCUMENT NOTICE: [SYNTHETIC DEMONSTRATION ARTIFACT] Synthetic engineering design specification created for TechPulse FY-26 Capstone evaluation. Illustrative architectural content.

1. Scope and Perception Architecture

Document ID: HLD-SYN-ADAS-003 | Revision: 3.0 | Target Domain: Autonomous Driving L2+
This high-level design governs the front-facing multi-modal perception pipeline, combining an 8-megapixel monocam vision pipeline and a 77 GHz millimetre-wave front radar sensor over Automotive Gigabit Ethernet (1000BASE-T1).

2. Adaptive AUTOSAR Execution Context

The fusion pipeline runs inside an Adaptive AUTOSAR platform on a multi-core SoC (Safety OS + POSIX Execution Management), communicating with the Vehicle Control Unit (VCU) across an AUTOSAR Classic gateway.

3. Software Components and SOME/IP Communication

Component Name: FrontCameraTracker_SWC
Description: Consumes optical object detection bounding boxes, 3D world pose estimates, and lane polygon coefficients.
Interfaces: Consumes SOME/IP Event *VisionObjectTracks* (50ms period, payload size 12 KB).
Outputs: *PPort_CameraTracks* providing object class (pedestrian, vehicle, cyclist) and lateral confidence.

Component Name: RadarClusterProcessor_SWC
Description: Filters radar reflection point clouds, calculates Doppler range rates, and generates clustered target vectors.
Interfaces: Consumes CAN-FD cluster frames; provides *PPort_RadarTargets* (20ms period, high longitudinal velocity accuracy).

Component Name: ObjectFusionArbitrator_SWC
Description: Implements Joint Probabilistic Data Association (JPDA) and track-to-track covariance intersection.
Outputs: *PPort_FusedTrackList* published to Chassis Gateway and ADAS Path Planner.

4. Forward Collision Warning (FCW) Functional Flow

Sequence FF-ADAS-03: Lead Vehicle Braking Alert Flow
Step 1: *FrontCameraTracker_SWC* classifies lead target vehicle at 65 meters ahead.
Step 2: *RadarClusterProcessor_SWC* detects relative range rate closing at -18 m/s.
Step 3: *ObjectFusionArbitrator_SWC* computes Time-to-Collision (TTC): *TTC* = *Range* / *Relative_Speed*. TTC evaluates to 2.1 seconds.
Step 4: When TTC falls below 2.4 seconds, an acoustic and visual FCW alert is triggered via Instrument Cluster SWC.
Step 5: If TTC drops below 1.4 seconds without driver intervention, an Autonomous Emergency Braking (AEB) command is transmitted to the Brake-by-Wire ECU.

5. Sensor Occlusion & Graceful Degradation Constraints

- **Degradation Policy:** If heavy snowfall or mud obscures the optical camera lens, *FrontCameraTracker_SWC* raises blind flag.
- **Fallback State:** *ObjectFusionArbitrator_SWC* enters Radar-Only Degraded Mode with increased headway margin (+20m).
- **ASIL Allocation:** FCW alert path is ASIL B; AEB deceleration trigger path is ASIL D.
- **End-to-End Latency:** Raw radar/camera ingress to fused output broadcast must not exceed 45 milliseconds.